

Q1 - 24 June - Shift 1

Two identical thin biconvex lenses of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____ cm.

Space for your notes:

Q2 - 24 June - Shift 2

A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is _____ cm.

Space for your notes:

Q3 - 25 June - Shift 1

A light wave travelling linearly in a medium of dielectric constant 4, incident on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be (Given : relative permeability of medium $\mu_r = 1$)

- (A) 10° (B) 20°
(C) 30° (D) 60°

Space for your notes:

Q4 - 25 June - Shift 1

The difference of speed of light in the two media A and B ($v_A - v_B$) is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum $c = 3 \times 10^8$ ms⁻¹)

- (A) 1.303
- (B) 1.318
- (C) 1.13
- (D) 0.12

Space for your notes:

Q5 - 25 June - Shift 2

A light whose electric field vectors are completely removed by using a good Polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

- (A) Reflected and refracted rays will be perpendicular to each other
- (B) Wave will propagate along the surface of prism
- (C) No refraction, and there will be total reflection of light.
- (D) No reflection and there will be total transmission of light.

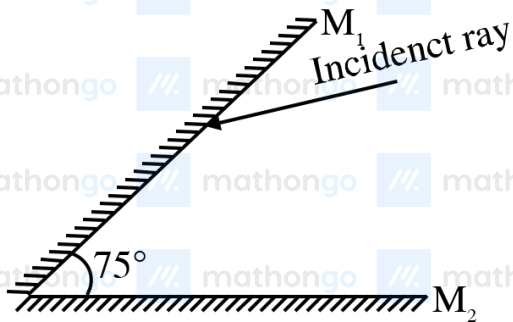
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Q6 - 26 June - Shift 1

Questions

MathonGo

A light ray is incident, at an incident angle θ_1 , on the system of two plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30° . The total deviation of the ray will be degree.

*Space for your notes:***Q7 - 26 June - Shift 2**

A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}$ m. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi$ m². The value of x is _____.

*Space for your notes:***Q8 - 27 June - Shift 1**

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Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be :

(A) $\sin^{-1}(\sqrt{n})$ (B) $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

(C) $\sin^{-1}(\sqrt{2n})$ (D) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

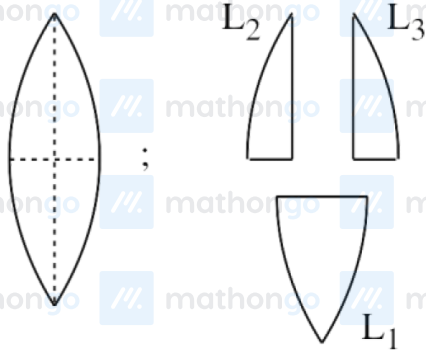
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Q9 - 27 June - Shift 2

Questions

MathonGo

A convex lens has power P . It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figure). Choose the incorrect option for the reported pieces.



(A) Power of $L_1 = \frac{P}{2}$

(B) Power of $L_2 = \frac{P}{2}$

(C) Power of $L_3 = \frac{P}{2}$

(D) Power of $L_1 = P$

*Space for your notes:***Q10 - 28 June - Shift 1**

The refracting angle of a prism is A and refractive index of the material of the prism is $\cot(A/2)$. Then the angle of minimum deviation will be -

(A) $180 - 2A$ (B) $90 - A$

(C) $180 + 2A$ (D) $180 - 3A$

*Space for your notes:***Q11 - 28 June - Shift 1**

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The aperture of the objective is 24.4 cm. The resolving power of this telescope. If a light of wavelength $2440 \text{ \AA}$ is used to see the object will be

- (A) 8.1×10^6 (B) 10.0×10^7
(C) 8.2×10^5 (D) 1.0×10^{-8}

Q12 - 29 June - Shift 1

A parallel beam of light is allowed to fall on a transparent spherical globe of diameter 30 cm and refractive index 1.5. The distance from the centre of the globe at which the beam of light can converge is _____ mm.

Q13 - 29 June - Shift 2

The speed of light in media 'A' and 'B' are $2.0 \times 10^{10} \text{ cm/s}$ and $1.5 \times 10^{10} \text{ cm/s}$ respectively. A ray of light enters from the medium B to A at an incident angle ' θ '. If the ray suffers total internal reflection, then

- (A) $\theta = \sin^{-1}\left(\frac{3}{4}\right)$ (B) $\theta > \sin^{-1}\left(\frac{2}{3}\right)$
(C) $\theta < \sin^{-1}\left(\frac{3}{4}\right)$ (D) $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

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Answer Key

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Q1 (10)

Q2 (12)

Q3 (D)

Q4 (C)

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Q5 (D)

Q6 (210)

Q7 (9)

Q8 (D)

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Q9 (A)

Q10 (A)

Q11 (C)

Q12 (225)

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Q13 (D)

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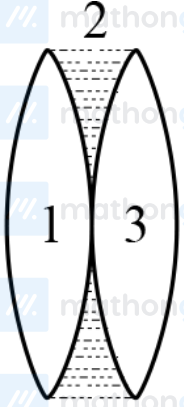
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$$\frac{1}{f_1} = \frac{1}{15} = \left(\frac{3}{2} - 1 \right) \left[\frac{2}{R} \right]$$

$$\frac{1}{R} = \frac{1}{15}$$

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$= \frac{1}{15} + \left(\frac{5}{4} - 1 \right) \left[\frac{-2}{R} \right] + \frac{1}{15}$$

$$= \frac{1}{15} - \frac{1}{30} + \frac{1}{15}$$

$$= \frac{2 - 1 + 2}{30}$$

$$= \frac{3}{30} = \frac{1}{10}$$

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Q2 (12)

$$\ell = t \sin i \left[1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}} \right]$$

$$\Rightarrow 4\sqrt{3} = t \sin 60^\circ \left[1 - \frac{\cos 60^\circ}{\sqrt{3 - \frac{3}{4}}} \right]$$

Q3 (D)

For total internal reflection, $i > \theta_c$

$$\Rightarrow \sin i > \sin \theta_c$$

$$\Rightarrow \sin i > \frac{\mu_R}{\mu_D} \dots\dots\dots(1)$$

$$\text{Also } \mu = \sqrt{\mu_r \epsilon_r}$$

$$\frac{\mu_R}{\mu_D} = \frac{\sqrt{1 \times 1}}{\sqrt{4 \times 1}} = \frac{1}{2}$$

$$\text{From (1), } \sin i > \frac{1}{2} \Rightarrow i > 30^\circ, i = 60^\circ$$

Q4 (C)

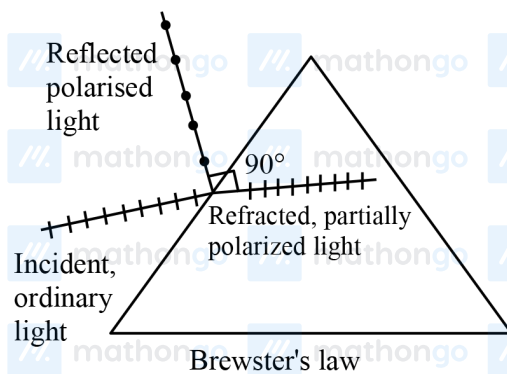
$$v = \frac{c}{\mu}$$

$$\Rightarrow v_B = \frac{3 \times 10^8}{1.47} = 2.04 \times 10^8 = 20.4 \times 10^7 \text{ m/s}$$

$$\therefore v_A - v_B = 2.6 \times 10^7 \text{ m/s}$$

$$\therefore v_A = (20.4 + 2.6) \times 10^7 = 23 \times 10^7 \text{ m/s}$$

$$\therefore \frac{\mu_B}{\mu_A} = \frac{v_A}{v_B} = \frac{23 \times 10^7}{20.4 \times 10^7} = 1.13$$

Q5 (D)

But as the incident light electric field vectors are completely removed so there will be no reflection and there will be total transmission of light, explained by an experiment in NCERT.

[Reference NCERT Part-2 Pg-380, (A special case of total transmission)]

Note : Since direction of polarization is not mentioned hence most suitable option (D)

corresponding to case in which electric field is absent perpendicular to plane consisting incident and normal.

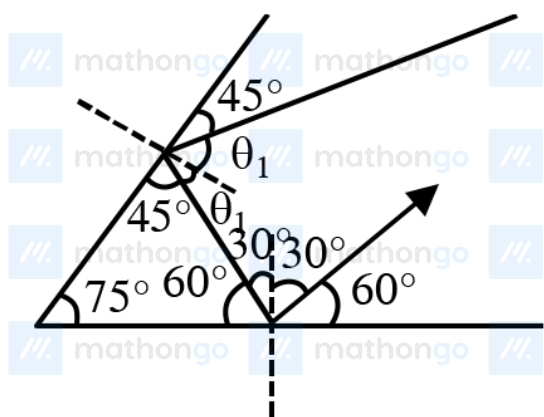
Q6 (210)

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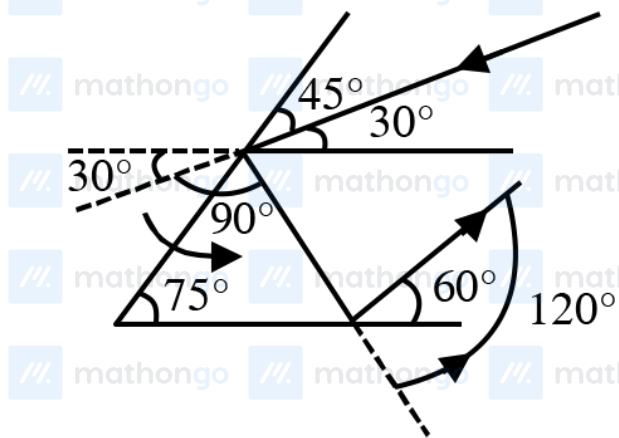
$$\delta_{\text{total}} = 360^\circ - 2\theta$$

$$= 360^\circ - 2 \times 75^\circ$$

$$\delta_{\text{total}} = 210^\circ$$



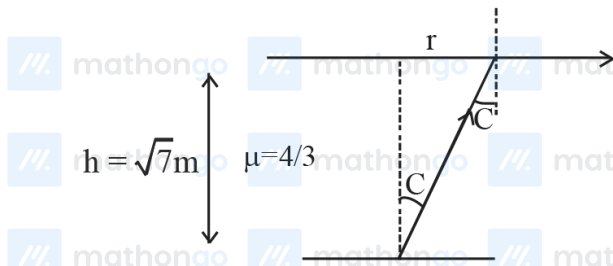
$$\theta_1 = 45^\circ$$



$$\delta = 120^\circ + 90^\circ = 210^\circ$$

Q7 (9)

C : Criticle angle



$$\tan C = \frac{r}{h}$$

$$r = h \tan C$$

$$\sin C = \frac{1}{\mu} = \frac{3}{4}$$

$$\tan C = \frac{3}{\sqrt{7}}$$

$$r = \sqrt{7} \times \frac{3}{\sqrt{7}} = 3$$

$$\text{Area of surface} = \pi r^2 = 9\pi\text{m}^2$$

Q8 (D)

$$13.6 \left(\frac{1}{1^2} - \frac{1}{n^2} \right) = 10.2$$

$$n = 2$$

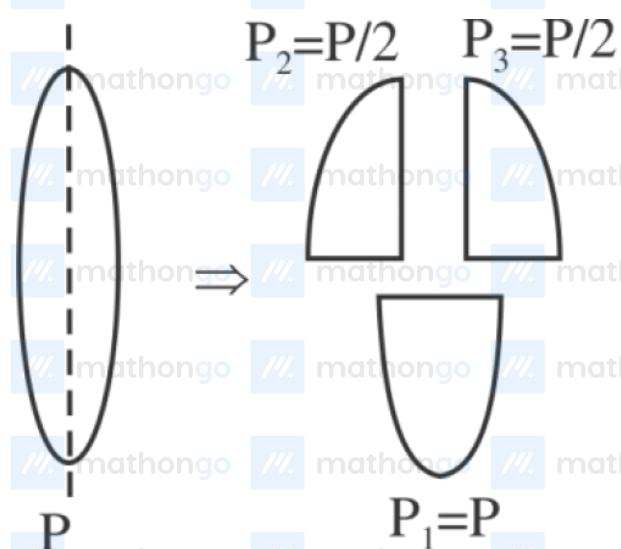
$$L_i = \frac{h}{2\pi} \times 1$$

$$L_F = \frac{2h}{2\pi}$$

$$\Delta L = L_F - L_i = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34}}{2 \times \frac{22}{7}}$$

$$= 1.05 \times 10^{-34} \text{ J-s}$$

Q9 (A)



Q10 (A)

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin \frac{A}{2}}$$

$$\mu = \cot \frac{A}{2}$$

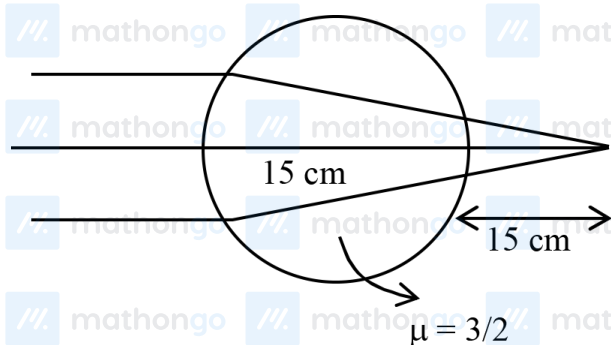
$$\Rightarrow \sin\left(\frac{A + \delta_m}{2}\right) = \cos \frac{A}{2}$$

$$\delta_m = 180 - 2A$$

Q11 (C)

$$R.P = \frac{d}{1.22\lambda} = \frac{24.4 \times 10^{-2}}{1.22 \times 2440 \times 10^{-10}} = 8.2 \times 10^5$$

Q12 (225)



$$\frac{3}{2} - \frac{1}{\infty} = \frac{3}{2} - \frac{1}{15}$$

$$\frac{3}{2V} = \frac{1}{30}$$

$$V = 45 \text{ cm}$$

$$\frac{1}{V} - \frac{3}{2} = \frac{1 - \frac{3}{2}}{\frac{2}{15}}$$

$$\frac{1}{V} - \frac{1}{10} = \frac{1}{30}$$

$$\frac{1}{V} = \frac{1}{10} + \frac{1}{30} = \frac{4}{30}$$

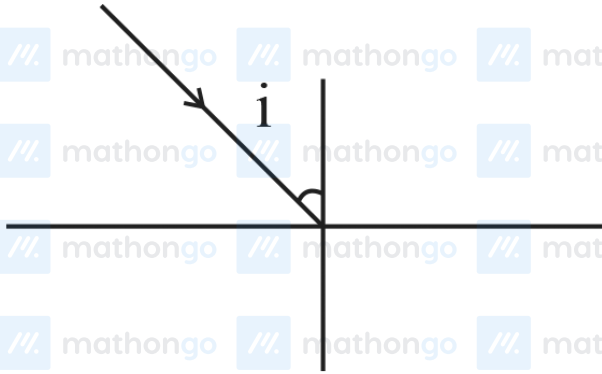
$$V = 7.5$$

$$V = 22.5$$

$$v = 225 \text{ mm}$$

Q13 (D)

$$\sin i_c = \frac{n_r}{n_d} = \frac{C_d}{C_r} = \frac{1.5 \times 10^{10}}{2 \times 10^{10}}$$



$$\sin i_c = \frac{3}{4}$$

$$i_c = \sin^{-1} \left(\frac{3}{4} \right)$$

for TIR $\theta > i_c$

$$\theta > \sin^{-1} \left(\frac{3}{4} \right)$$